

Amazon Product Review Analysis

How we used a Text CNN to predict Amazon
Fire Stick review sentiment from keywords.

Group 14: We KANT but We Can

DS 4002

Group Leaders: Alessia Natalucci, Nico Montoya
Group Members: Katie Lai, Tiannah Griffith

Project Details

Motivation

- 90%+ shoppers rely on reviews
- Be able to distinguish positive and negative amazon reviews for a product based on keywords.

Hypothesis:

- A text CNN can classify Amazon reviews as positive or negative with at least 80% accuracy.

Research Question:

- Can we run a model that can classify a product review as positive or negative based solely on text?

Modeling approach:

- Text CNN

Goal

- To be able to classify key phrases as negative or positive



Data Acquisition

Acquisition: Dataset created by UCSD. There were no licensing or ethical concerns.

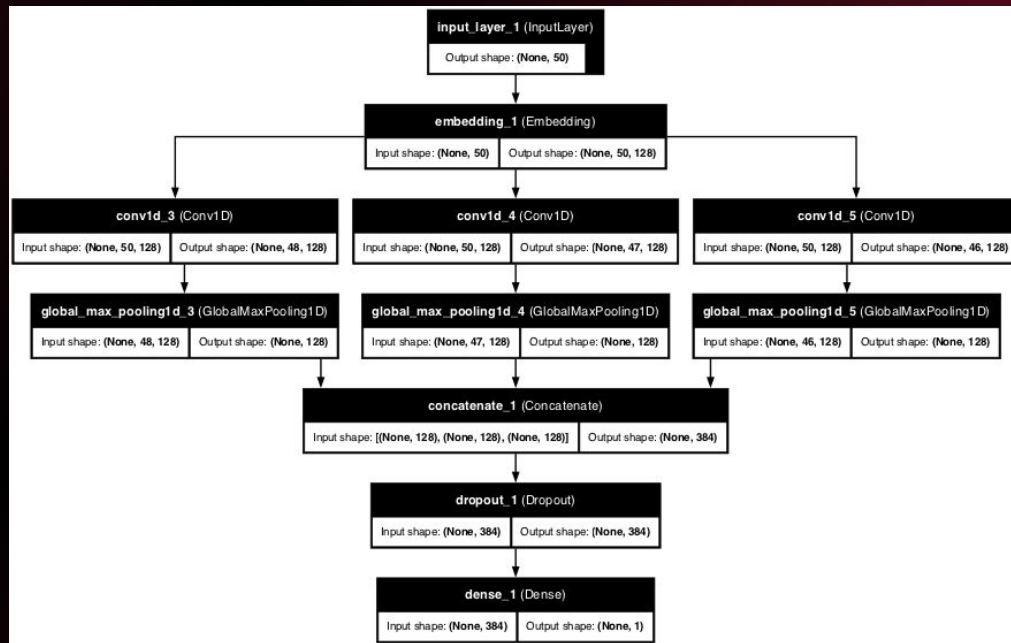
The original dataset had over 40 million rows and over a million products so the dataset was cleaned down to 1 product: the Amazon Firestick. The data was further cleaned to include only these columns.

Format: Text-based reviews concatenating the title and text columns into one text block which can be applied to the CNN.

Column	Description
rating	1 to 5 star rating of the product
title	Title of the review
text	Contents of the review
asin	Product ID code
timestamp	Year that the review was published
verified_purchase	Verification by Amazon that the reviewer purchased the product

Analysis Plan

1. Data cleaning
2. EDA
3. Tokenize and label data
4. Training and model application
5. Evaluation metrics
6. Compare evaluation metrics of stratified, non-stratified, and balanced data splits



Tricky Analysis Decisions

We dropped three star reviews

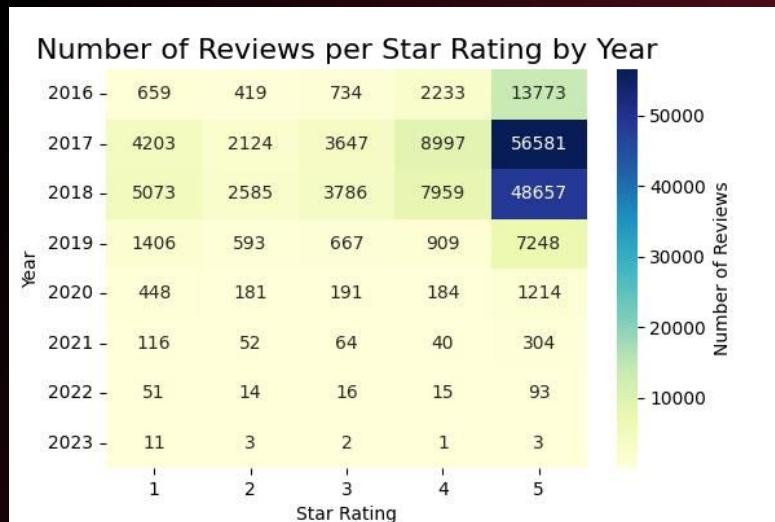
We ran three versions of the model to test accuracy

- Model with data stratification
- Model without data stratification model
- Model with balanced split of data

Bias and Uncertainty Validation

The main bias we attempted to correct was the class imbalance. After running the model we found the class distribution to be ~90% positive reviews.

We ran the three models and found that based on accuracy, the non-stratified split had the highest, then the stratified split and then the balanced split



Results and Conclusions

6

Cross entropy loss

Stratified: 0.35

Non-Stratified: 0.06

Balanced: 0.49

Non-Stratified: Accuracy ≈ 0.99 ; Precision ≈ 0.99 ; Recall ≈ 0.99 ; F1 ≈ 0.99

Stratified: Accuracy ≈ 0.96 ; Precision ≈ 0.97 ; Recall ≈ 0.98 ; F1 ≈ 0.98

7

Balanced: Accuracy ≈ 0.93 ; Precision ≈ 0.93 ; Recall ≈ 0.93 ; F1 ≈ 0.93

Conclusion: A CNN model was able to classify positive or negative reviews with at least 80% accuracy. The stratified model provides the most reliable and generalizable performance.

Next Steps...

- Expand dataset to include more products and review categories
- Incorporate star rating prediction (multi-class classification)
- Experiment with more advanced models
- Implement hyperparameter tuning to further optimize performance
- Deploy model as a real-time review classification tool
- Analyze key phrases driving positive vs. negative predictions

References

[1] "Amazon Reviews'23," amazon-reviews-2023.github.io. <https://amazon-reviews-2023.github.io/>

[2] R. Nandagopal, A. Jayakumar, and G. Manokaran, "Impact of Online Reviews on Consumer Purchasing Decisions."

Github: <https://github.com/anatalucci/DS4002-Project1.git>

THANK YOU



Questions?

We KANT, but we can